Literature Status for arXiv:1311.6584: The Planar (B)-Conjecture for Even Log-Concave Measures

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Status and source question

This is a `literature_already_answered (full planar case)` packet. In "Examples and open questions," on PDF page 2 of [1], Livne Bar-on writes:

> The (B)-conjecture for general centrally-symmetric log-concave measures is not settled yet, even in two dimensions. It is also of interest to generalize the method of this paper to higher dimensions.

Here the (B)-conjecture says that for every even log-concave measure μ and every origin-symmetric convex body K,

$$t \longmapsto \mu(e^t K)$$

is log-concave.

Later answer

Saroglou proves the precise planar statement in [2, Corollary 3.3, PDF p. 5]:

> The log-Brunn–Minkowski inequality and the B-conjecture hold in the plane, for any even log-concave density.

The route is explicit. Theorem 3.1 transfers the log-Brunn–Minkowski inequality from Lebesgue measure to every even log-concave density; Corollary 3.2 derives the (B)-conjecture in the same dimension; and the known planar log-Brunn–Minkowski theorem gives Corollary 3.3.

The supporting paper explicitly recognizes the source. Its introduction states that the planar uniform-measure case had been proved independently in [1], and its bibliography identifies that reference as arXiv:1311.6584. This is therefore an exact later-literature answer, not an agent-created proof. Measures supported on a proper linear subspace reduce to their support (dimension zero or one), so the density formulation covers the substantive full-dimensional planar case without leaving a singular-measure gap.

Scope and search bounds

The answer is complete for the source's explicit "even in two dimensions" subproblem. It does not settle the higher-dimensional (B)-conjecture for all even log-concave measures.

The run's cheap indexes were searched for the source arXiv id, title, and core terms. A bounded arXiv/web search for the exact planar statement found arXiv:1409.4346 and later surveys confirming the same status. The decisive paper was checked in source form at its abstract, introduction, Corollaries 3.2–3.3, and bibliography entry for [1].

References

[1] A. Livne Bar-on, *The (B) conjecture for uniform measures in the plane*, arXiv:1311.6584 (2013), arXiv:1311.6584.

[2] C. Saroglou, *More on logarithmic sums of convex bodies*, arXiv:1409.4346 (2014); Mathematika **62** (2016), 818–841, arXiv:1409.4346.